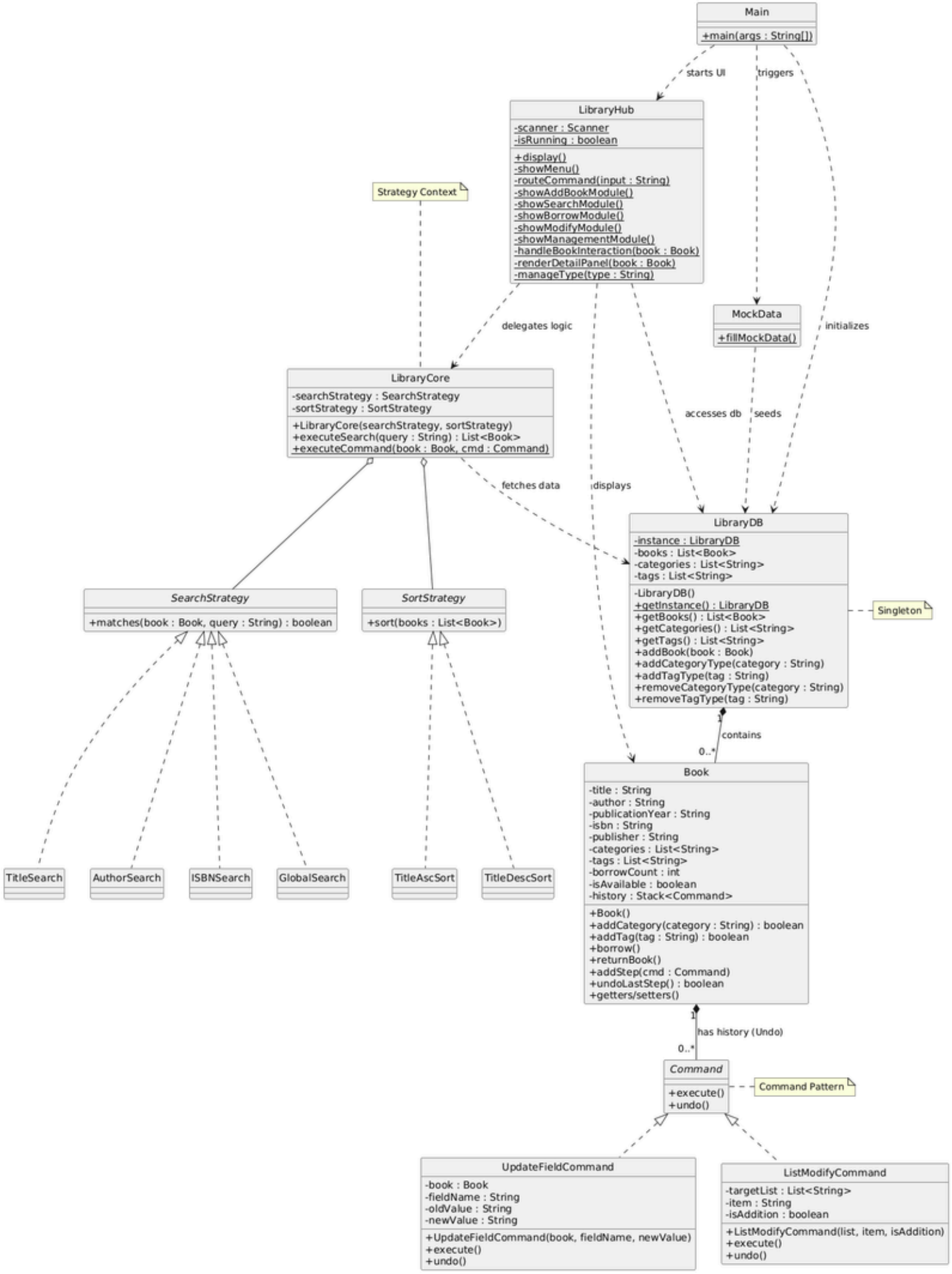
Main
+main(args : String[])
starts UI
triggers
initializes
LibraryHub
-scanner : Scanner
-isRunning : boolean
+display()
-showMenu()
-routeCommand(input : String)
-showAddBookModule()
-showSearchModule()
-showBorrowModule()
-showModifyModule()
-showManagementModule()
-handleBookInteraction(book : Book)
-renderDetailPanel(book : Book)
-manageType(type : String)
Strategy Context
MockData
+fillMockData()
delegates logic
accesses db
seeds
LibraryCore
-searchStrategy : SearchStrategy
-sortStrategy : SortStrategy
+LibraryCore(searchStrategy, sortStrategy)
+executeSearch(query : String) : List<Book>
+executeCommand(book : Book, cmd : Command)
fetches data
displays
LibraryDB
-instance : LibraryDB
-books : List<Book>
-categories : List<String>
-tags : List<String>
-LibraryDB()
+getInstance() : LibraryDB
+getBooks() : List<Book>
+getCategories() : List<String>
+getTags() : List<String>
+addBook(book : Book)
+addCategoryType(category : String)
+addTagType(tag : String)
+removeCategoryType(category : String)
+removeTagType(tag : String)
Singleton
SearchStrategy
+matches(book : Book, query : String) : boolean
SortStrategy
+sort(books : List<Book>)
1
contains
0..*
Book
-title : String
-author : String
-publicationYear : String
-isbn : String
-publisher : String
-categories : List<String>
-tags : List<String>
-borrowCount : int
-isAvailable : boolean
-history : Stack<Command>
+Book()
+addCategory(category : String) : boolean
+addTag(tag : String) : boolean
+borrow()
+returnBook()
+addStep(cmd : Command)
+undoLastStep() : boolean
+getters/setters()
TitleSearch
AuthorSearch
ISBNSearch
GlobalSearch
TitleAscSort
TitleDescSort
1
has history (Undo)
0..*
Command
+execute()
+undo()
Command Pattern
UpdateFieldCommand
-book : Book
-fieldName : String
-oldValue : String
-newValue : String
+UpdateFieldCommand(book, fieldName, newValue)
+execute()
+undo()
ListModifyCommand
-targetList : List<String>
-item : String
-isAddition : boolean
+ListModifyCommand(list, item, isAddition)
+execute()
+undo()